Topic 5 Test

ATTEMPT SCORE

37 / 40

01 Matching 5 / 6

Matching.

Match the types of heat transfer to the examples.

A campfire warms the side of your body that's facing the fire.	A	radiation ✔
Your chair is stealing your thermal energy! Taking it right from you!	B	conduction ✔
The surface of your planet is warmed by the Sun.	C	radiation ✔
Standing next to a river of lava is dangerous because the air is so hot.	D	convection ✔
Yeowch! Your lab partner Throckmorton spilled boiling water on you! Go run that under water in the sink for five minutes!	E	convection ✔
The water boiling in Mx. Casanova's electric kettle.	F	convection ✔

conduction convection radiation

02 Short Answer/Essay

4 / 4

Short answer.

Jiwon is baking cookies for his science fair project in a conventional oven (a singular heating element at the bottom of the oven). When he pulls out the first batch of cookies, he sees that the cookies from the top rack are perfectly baked, but the cookies from the bottom rack are burned.

What happened, and how would using a convection oven help Jiwon out?

As stated, convectional ovens are ovens that have only one heating element that is at the bottom. So, in this case, what happened is that the cookies on the bottom row got more heat than the cookies on the top row. This was because of convection currents. See, When air heats up it rises, when the air cools it cools down and gets sent to the bottom again. But since the cookies on the bottom row were super close to the heating element, the air couldn't cool down. So the cookies kept on baking until it was burned. The cookies on the top row got just enough heat to be perfect. Convection ovens would be a better choice for Jiwon because it helps spread the air evenly. So the convection current is maintained. This is what happened while Jiwon was baking cookies and why convection ovens are a better choice.

150 / 150 Word Limit

03 Matching

10 / 10

Matching.

Match the example to the correct vocabulary term. *Not all examples will be used!*

Term		Example
Absolute Zero	A	An ice cube so cold that there's no molecular movement whatsoever. ✔
Conservation of Energy	B	Energy cannot be created nor destroyed, only transformed. ✔
Heat	C	Holding a warm cup of hot chocolate warms your hands. ✔
Temperature	D	It's 54°F outside! ✔
Thermal Energy	E	A cup of tea has a whole lot more of this than a refreshing cold soda. ✔

Mx. Casanova's mini fridge is mostly made of plastic.

04 Multiple Choice

0 / 2

Choose the best answer.

Boiling water and steam can both have a temperature of 100°C, but steam has more thermal energy. Why?

- Thermal energy is required for a substance to change states. ✔
- Steam has more chemical energy than liquid water.
- Witchcraft.
- Liquid water is a better conductor of heat than steam. ✕

05 True/False 2 / 2

True or false.

Microwaving two burritos at once will take the same amount of time as microwaving one burrito.

- True
- False ✔

Student's Correction

Microwaving two burritos needs a lot more thermal energy than microwaving one burrito because 2 burritos have more particles than a single burrito. So heating up the 2 burrotos will take longer.

Correction Guidelines

06 Multiple Choice 2 / 2

Choose the best answer.

Which statement best describes the relationship between molecular movement and temperature?

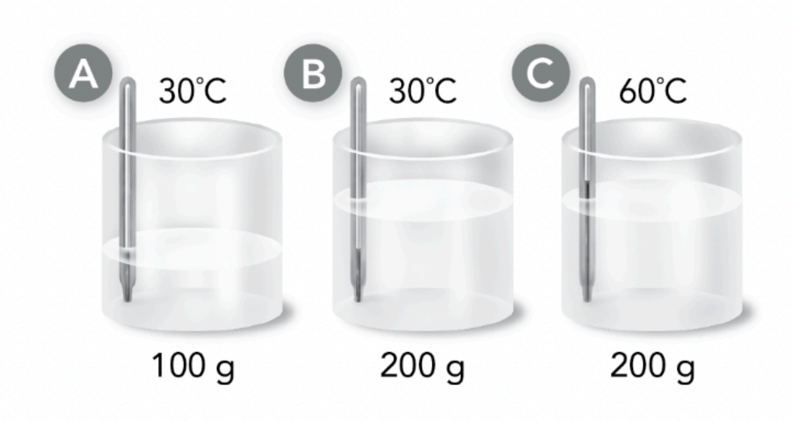
- the faster the movement, the higher the temperature ✔
- there is no relationship between molecular movement and temperature
- fluids are hot because the circulation of molecules creates thermal energy
- molecular movement and temperature are both characteristics of radiation

07 Multiple Choice

2 / 2

Choose the best answer.

Which of the following beakers of water has the **least** thermal energy?



- ◉ Beaker A ✓
- ○ Beaker B
- ○ Beaker C
- ○ Beakers A and B have the same amount of thermal energy.

08 Fill in the Blank Drag and Drop

4 / 4

Fill in the blanks.

Complete the following statements:

1. Mx. Casanova's kettle transforms 1 electricity ✔ into 2 thermal energy ✔ to heat water for their tea.

2. Rubbing your hands together transforms 3 kinetic energy ✔ into 4 thermal energy ✔ .

3. The motor in your mom's Tesla transforms 5 electricity ✔ into 6 kinetic energy ✔ so that you can get to school.

4. When Mx. Casanova lights a piece of paper on fire, the 7 chemical energy ✔ in the paper is transformed into 8 thermal energy ✔ .

chemical energy | electricity | kinetic energy | thermal energy

09 Multiple Choice

2 / 2

Choose the best answer.

You are camping, and it is time to build a campfire. You gather logs and stack them loosely, and then you add small twigs and leaves to make the fire easy to light. As you sit by the fire, the logs burn bright in the darkness.

Which statement is true about the amount of energy transformed in the campfire?

- (selected) the amount of heat and light energy release equals the amount of chemical energy that is released as the logs burn ✔
- the fire is hotter than the logs were, so energy is gained in this transformation
- the amount of energy transformed is equal to zero because burning is a physical change
- the chemical energy in the logs is lost during the transformation

10 Matching

6 / 6

Matching.

Label the following materials as either insulators or conductors.

air	A	INSULATOR ✓
aluminum	B	CONDUCTOR ✓
ceramic	C	INSULATOR ✓
copper	D	CONDUCTOR ✓
steel	E	CONDUCTOR ✓
wool	F	INSULATOR ✓

INSULATOR CONDUCTOR